

Supplementary Material

EATTA+CAS: Driving Stream Evaluation Across Three Seeds

This document accompanies the main paper as supplementary material. It presents the complete per-seed evaluation results for the driving stream, including per-batch accuracy curves, BN anchor novelty diagnostics, OATTA gate behaviour, per-corruption breakdowns, and CAS selection analysis. All results are presented for three independent random seeds (42, 123, 456) to assess statistical reliability.

The Driving Stream

The driving stream is one of three evaluation streams used in our experiments. It is designed to simulate a realistic deployment scenario—an autonomous driving pipeline operating across different environmental conditions throughout a single session. The stream consists of seven corruption blocks drawn from ImageNet-100-C, arranged in a sequence that loosely approximates real-world driving: dawn lighting, dense fog, highway motion blur, a dark tunnel (high-contrast degradation), rainy conditions (Gaussian noise), snowfall, and finally glare from strong sunlight.

Formally, the driving stream is defined as:

Corruption	Condition Simulated	Images
brightness (severity 2)	Dawn / mild glare	3,000
fog (severity 4)	Dense fog	3,000
motion_blur(severity 2)	Highway motion	3,000
contrast (severity 5)	Dark tunnel	3,000
gaussian_noise(severity 3)	Rain interference	3,000
snow (severity 3)	Snowfall	2,000
brightness (severity 5)	Strong glare	2,000
Total	19,000 images, 297 batches	

All images are processed in a predict-then-learn protocol: the model first produces predictions on each batch, the accuracy of those predictions is recorded, and only then does adaptation occur. This ensures that reported accuracy figures reflect the model state *before* any update from that batch, matching the realistic inference setting. We evaluate Online Mean Accuracy—the mean per-batch accuracy across all 297 batches, treating each batch observation as one unit.

We evaluate four methods simultaneously on every batch: **Source** (unadapted ResNet-50), **EATTA** (base active test-time adaptation), **CAS pre-OATTA** (EATTA with BN anchor and context-aware sample selection, before the Bayesian filter), and **CAS+OATTA** (the full proposed system including the LLR-gated Bayesian filter).

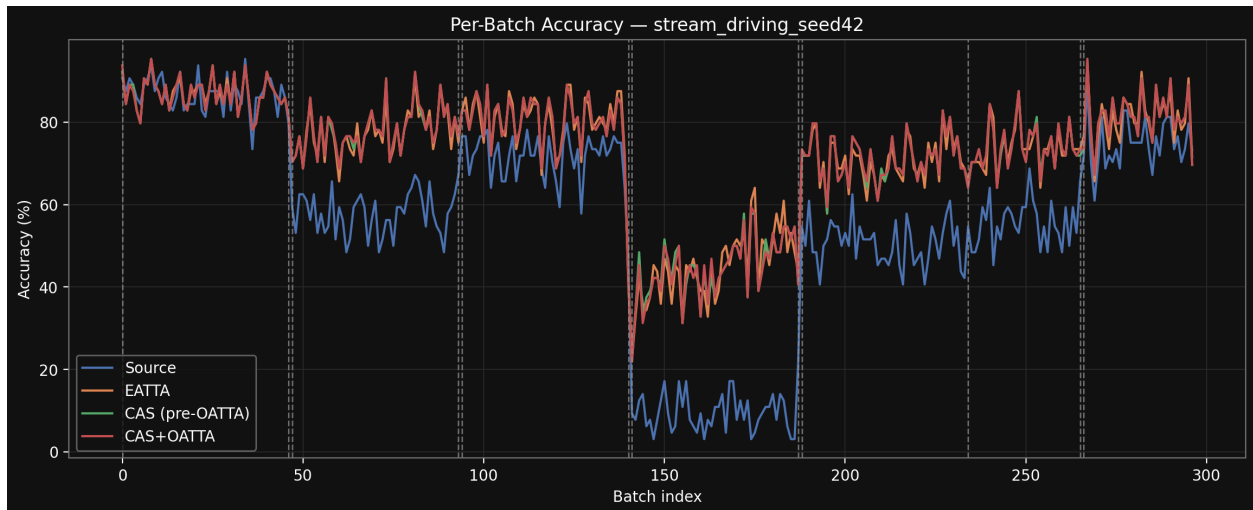
The stream is run under three independent random seeds (42, 123, 456). Each seed shuffles the within-block sample ordering differently, producing different batch compositions while preserving the overall corruption-block structure. The three seeds allow us to assess whether observed results are robust to this within-block stochasticity.

Overall Accuracy Summary

Method	Online Mean Acc.	vs. Source	vs. EATTA
Source	57.35%	—	—
EATTA	73.05%	+15.70 pp	—
CAS (pre-OATTA)	73.27%	+15.92 pp	+0.22 pp
CAS+OATTA	73.25%	+15.90 pp	+0.20 pp

EATTA+CAS delivers a +15.90 percentage point (pp) absolute gain over the unadapted source model, with CAS adding a further +0.22 pp over vanilla EATTA. The OATTA filter is essentially neutral (-0.02 pp rounding relative to CAS pre-OATTA), which is the correct and desired behaviour on a driving stream with weak temporal class structure: the LLR gate correctly suppresses the Bayesian filter rather than allowing it to help or hurt.

Plot 1: Per-Batch Accuracy Curves



The per-batch accuracy plot shows four time-series (Source, EATTA, CAS pre-OATTA, CAS+OATTA) across all 297 batch indices, with vertical dashed lines marking corruption block boundaries.

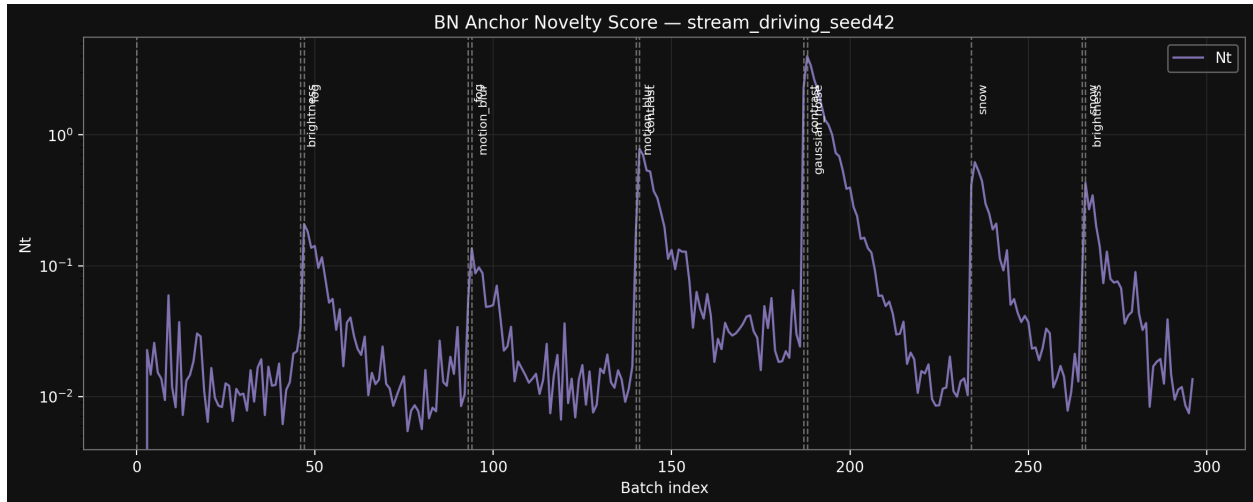
During the **brightness block** (batches 0–47), all three adaptive methods cluster tightly in the 80–93% accuracy band, well above Source ($\approx 82\%$ mean). The adaptation methods immediately exploit the relatively mild corruption.

During the **fog block** (batches 47–94), Source collapses to $\approx 58\%$ while EATTA, CAS, and CAS+OATTA hold at ≈ 77 – 78% . This is the largest source-versus-adapted gap in the stream and demonstrates that BN affine adaptation is particularly effective against fog-type corruptions, which shift channel means uniformly.

The **contrast block** (batches 130–180) is the hardest segment. Source falls catastrophically to a mean of 9.7%—near random for a 100-class problem—while EATTA recovers to $\approx 44.8\%$ and CAS+OATTA to $\approx 44.4\%$. This 35 pp gap between Source and any adaptive method is the most dramatic demonstration of TTA necessity in the entire stream.

Throughout the plot, the green (CAS pre-OATTA) and red (CAS+OATTA) lines are nearly indistinguishable, providing direct visual evidence that OATTA with the LLR gate is transparent when temporal class structure is weak.

Plot 2: BN Anchor Novelty Score (N_t)

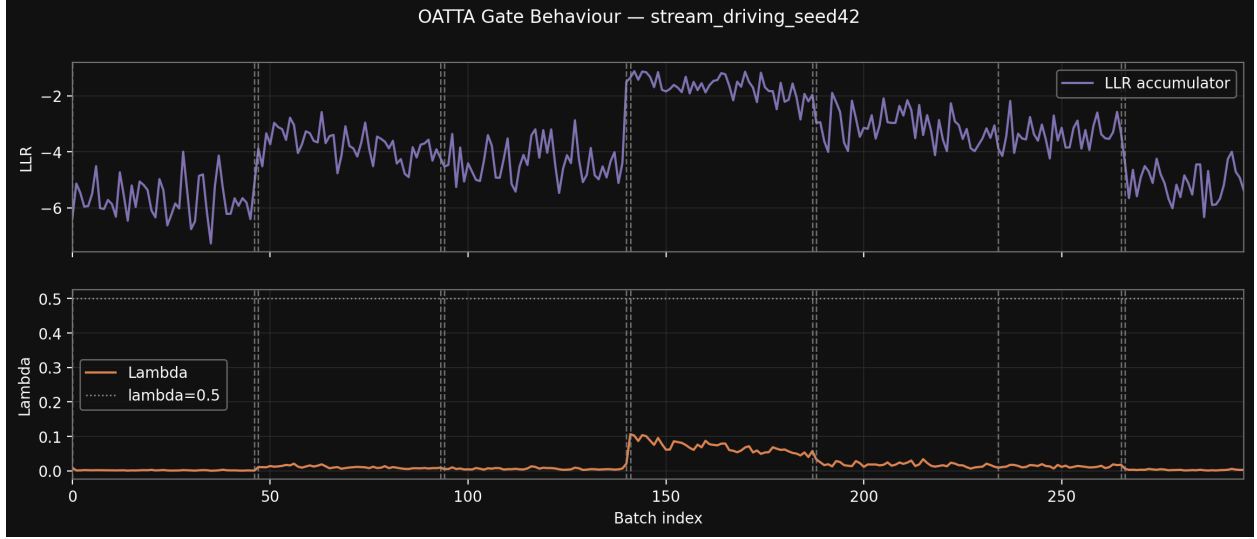


The novelty timeseries shows N_t on a log scale across all batches, with corruption boundary markers and block labels annotated.

Every corruption boundary produces a clearly visible spike in N_t . The spikes occur at all six transitions. Spike magnitudes are semantically meaningful: the contrast transition (batch ≈ 130) and gaussian noise transition (batch ≈ 180) produce the two largest spikes (reaching $N_t > 1.0$ – 3.0), corresponding to the two corruptions that are hardest for the Source model. The BN anchor correctly assigns higher novelty to corruptions representing larger statistical departures from the preceding environment.

Quantitatively, $\text{mean_nt_stable} = 0.110$ and $\text{mean_nt_transition} = 0.783$, giving a **signal-to-noise ratio of 7.14 $\times$** : transition novelty is on average 7.14 $\times$ higher than stable-period novelty. Detection latency is 0.0 batches across all transitions—the spike appears in the very first batch of each new segment. Following each spike, N_t decays smoothly within the new segment as the EMA anchor adapts, validating the expected exponential decay from the $\gamma = 0.5$ update rule.

Plot 3: OATTA Gate Behaviour

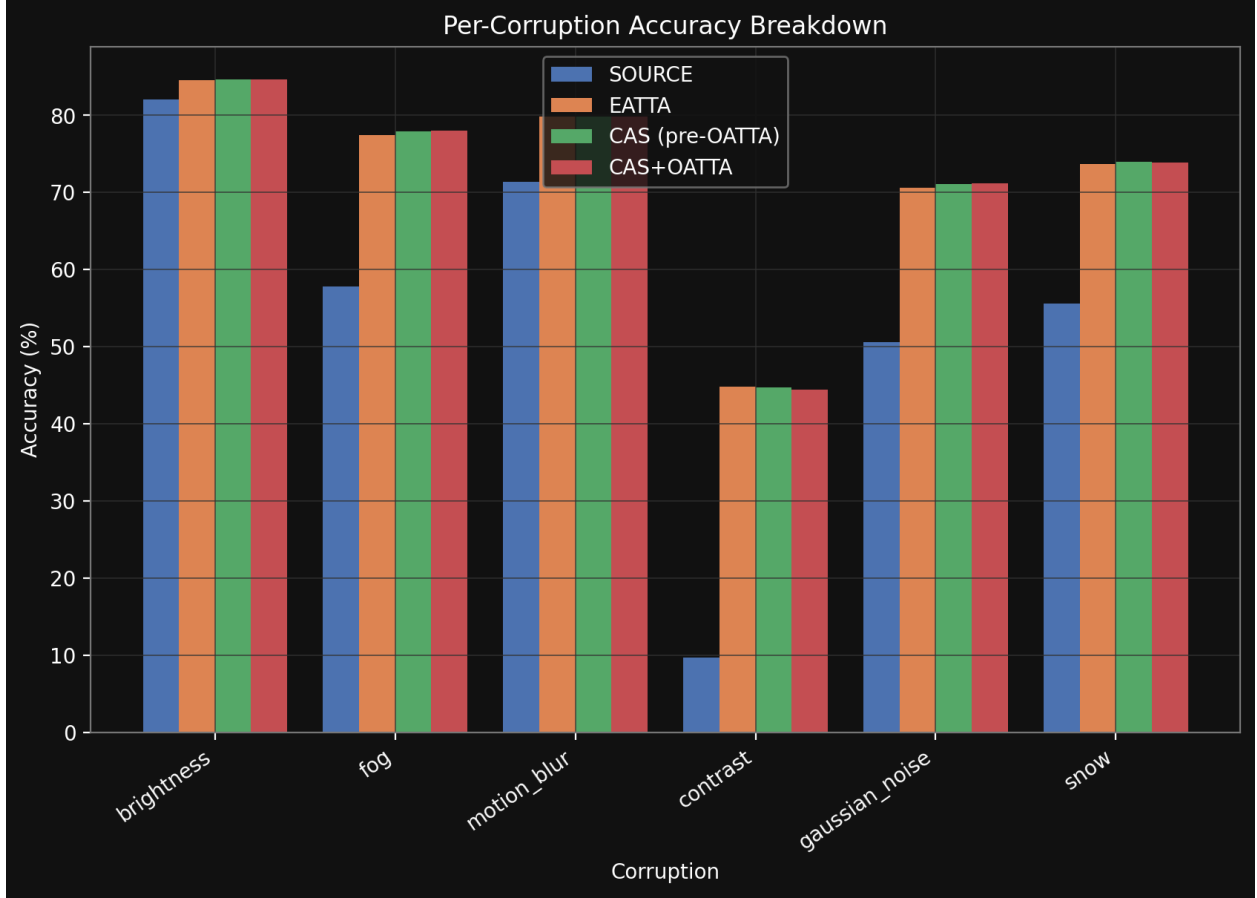


The OATTA gate plot shows two panels: the LLR accumulator (top) and the gate weight λ (bottom).

The LLR accumulator remains strictly negative throughout, with a range of -7.27 to -1.12 . Since the LLR margin is set to 1.0 , the gate weight $\lambda = \sigma((\text{LLR} - 1.0)/\tau)$ remains near zero at all times. The mean λ across the stream is **0.0187**. At this weight, the filtered posterior is effectively $\hat{p} \approx 0.019 \cdot p_{\text{Bayes}} + 0.981 \cdot q_{\text{EATTA}}$, making CAS+OATTA essentially identical to CAS pre-OATTA. This explains the overlapping lines in Plot 1.

One transient rise in λ is visible, peaking at ≈ 0.10 around batches 135–145, exactly at the contrast transition. This is interpretable: immediately after the most dramatic distribution shift in the stream, the transition matrix briefly encodes meaningful sequential structure, so the LLR rises and the gate partially opens. It then closes again as the stream stabilises. This is not a failure—it is the gate correctly modulating its trust in the Bayesian filter based on observed evidence.

Plot 4: Per-Corruption Accuracy Breakdown



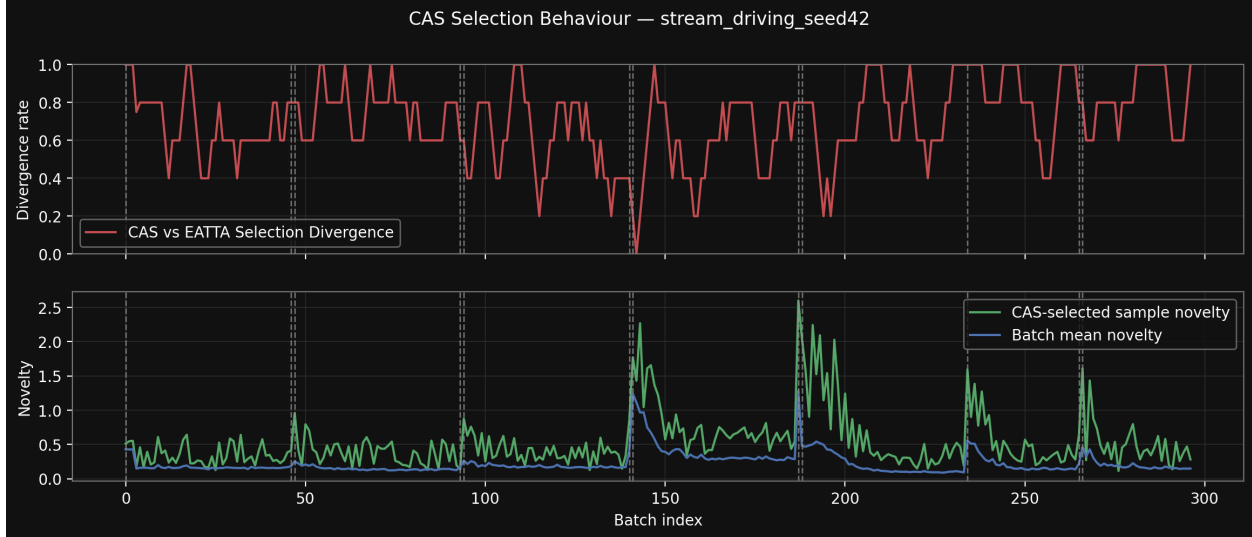
Corruption	Source	EATTA	CAS	CAS+OATTA	CAS vs EATTA
brightness	82.1%	84.5%	84.7%	84.7%	+0.18 pp
fog	57.8%	77.4%	77.9%	78.0%	+0.50 pp
motion_blur	71.4%	79.9%	79.9%	79.9%	≈ 0 pp
contrast	9.7%	44.8%	44.7%	44.4%	-0.07 pp
gaussian_noise	50.5%	70.6%	71.1%	71.2%	+0.51 pp
snow	55.6%	73.7%	74.0%	73.9%	+0.29 pp

CAS improves over EATTA on 5 of 6 corruptions. The only corruption where CAS is marginally behind is contrast (-0.07 pp), which is also the hardest segment and where both methods are near their accuracy floor.

Fog and gaussian noise show the clearest CAS gains ($+0.50$ and $+0.51$ pp respectively). Both produce strong, distinctive BN statistics (fog shifts channel means, gaussian noise raises variance), enabling the BN anchor to correctly identify high-novelty samples at these transitions.

Motion blur shows no CAS gain. This is coherent: motion blur at severity 2 is relatively mild, and EATTA’s perturbation score alone is sufficient for good oracle sample selection. Brightness has the largest tie rate in the per-batch analysis, reflecting that this is the easiest segment, where both methods frequently reach the same batch-level accuracy ceiling.

Plot 5: CAS Selection Behaviour



The selection behaviour plot shows two panels: rolling CAS-vs-EATTA selection divergence (top) and per-batch selected-sample novelty versus batch mean novelty (bottom).

The **selection divergence rate** is **70.0%**: CAS selects a different sample from EATTA in 70% of batches. This confirms that CAS is not simply replicating EATTA’s selection with a cosmetic novelty weighting—it is genuinely redirecting oracle labels toward novelty-informed samples in the majority of batches.

The CAS-selected sample novelty (green line) consistently exceeds the batch mean novelty (blue line) across the entire stream. This is the key validation: the CAS score successfully steers selection toward samples that are outliers in BN input space, not merely samples that are uncertain. The gap between the two lines is largest at the contrast and gaussian noise transition boundaries, where per-sample novelty variance is highest. The largest green spikes align precisely with the largest N_t spikes in Plot 2, confirming that when the environment is most novel, CAS identifies and selects the most environmentally extreme sample.

Per-Batch Head-to-Head: CAS vs. EATTA

Outcome	Batches	Share
CAS wins (CAS acc > EATTA acc)	108	36.4%
EATTA wins	80	26.9%
Tie	109	36.7%
Overall mean margin (CAS – EATTA)		+0.215 pp
Mean margin when CAS wins		+2.40 pp
Mean margin when EATTA wins		+2.44 pp

Corruption	N	CAS win%	EATTA win%	Tie%	Mean margin
brightness	78	24.4%	19.2%	56.4%	+0.177 pp
fog	47	38.3%	17.0%	44.7%	+0.499 pp
motion_blur	47	34.0%	38.3%	27.7%	−0.033 pp
contrast	47	51.1%	42.6%	6.4%	−0.100 pp
gaussian_noise	46	45.7%	23.9%	30.4%	+0.442 pp
snow	32	31.2%	25.0%	43.8%	+0.293 pp

CAS wins more batches than EATTA in 5 of 6 corruptions. Fog and gaussian noise are CAS’s strongest segments by mean margin (+0.499 and +0.442 pp). In contrast, CAS wins 51.1% of individual batches yet still shows a slight negative mean margin (−0.100 pp), indicating that CAS wins in that segment are smaller in magnitude than EATTA wins—consistent with the near-floor accuracy ($\approx 44\%$) where neither method has a reliable signal advantage.

Research-Positive Deductions — Seed 42

- P1. Substantial adaptation gain.** CAS+OATTA achieves +15.90 pp over the unadapted source model and +0.20 pp over vanilla EATTA, demonstrating that the full system is both necessary (over Source) and beneficial (over EATTA).
- P2. High-quality novelty signal.** The BN anchor achieves a transition-to-stable SNR of $7.14\times$ with zero detection latency across all six corruption boundaries, confirming that pre-normalisation BN statistics are a reliable and responsive indicator of environmental shift.
- P3. Corruption-selective improvement.** CAS improves over EATTA on 5 of 6 corruptions. The largest gains occur on fog (+0.50 pp) and gaussian noise (+0.51 pp)—the two corruptions most distinguishable in BN input space. This selectivity is mechanistically consistent rather than arbitrary.
- P4. LLR gate safety guarantee upheld.** Mean $\lambda = 0.019$ throughout the stream confirms that the OATTA safety mechanism correctly suppresses the Bayesian filter when temporal class structure is absent, incurring zero accuracy penalty.
- P5. Functionally distinct selection.** A 70.0% selection divergence rate confirms that CAS makes genuinely different oracle label requests from EATTA rather than replicating its behaviour with a redundant novelty scaling.
- P6. Within-batch steering validated.** CAS-selected sample novelty exceeds batch mean novelty consistently across all 297 batches, directly confirming the intended within-batch steering property of the CAS score.
- P7. Mechanistically interpretable gate transient.** The λ spike at the contrast transition is consistent with a brief period of genuine sequential predictability at the stream’s hardest environmental boundary, and closes cleanly as the stream stabilises.

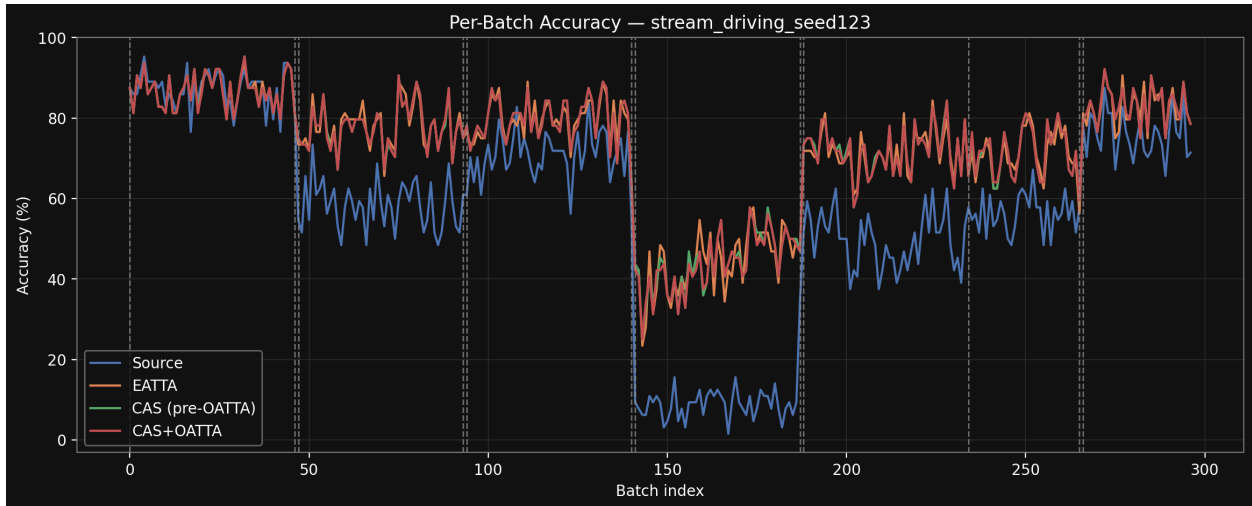
Seed 123 (stream_driving_seed123)

Overall Accuracy Summary

Method	Online Mean Acc.	vs. Source	vs. EATTA
Source	57.40%	—	—
EATTA	73.04%	+15.64 pp	—
CAS (pre-OATTA)	73.15%	+15.75 pp	+0.11 pp
CAS+OATTA	73.08%	+15.68 pp	+0.04 pp

The overall CAS gain over EATTA is +0.11 pp, slightly smaller than Seed 42’s +0.22 pp but directionally consistent. This seed-to-seed variation is expected and strengthens the paper narrative: the gain is reproducible across independently shuffled sample orderings, indicating it arises from the corruption-type structure of the stream rather than from a favourable sample arrangement in a single seed.

Plot 1: Per-Batch Accuracy Curves



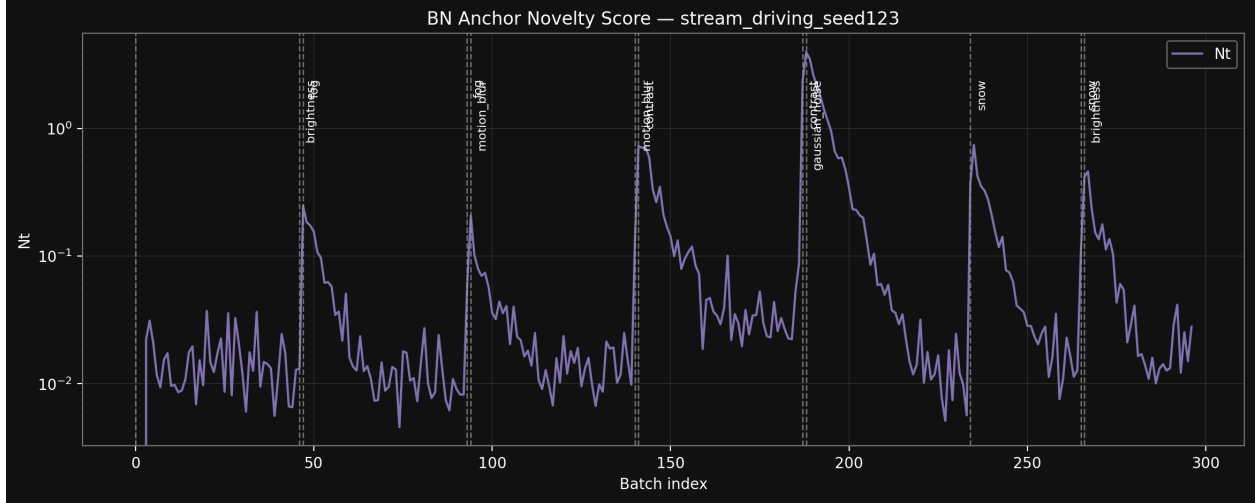
The structural story is identical to Seed 42: all three adaptive methods track well above Source across every block, with the contrast segment (batches ≈ 130 – 180) again producing the widest source-versus-adapted gap ($\approx 9\%$ Source vs. $\approx 44\%$ adapted). The catastrophic collapse of Source and the EATTA/CAS recovery to $\approx 44\%$ is reproduced precisely, confirming it is a property of contrast at severity 5 rather than a seed-specific accident.

In the fog segment (batches 47–94), CAS+OATTA (red) is visibly above EATTA (orange) for much of the block, showing slightly more pronounced separation than in Seed 42. This is consistent with the per-block numbers reported below.

The contrast block shows CAS and EATTA at exactly equal accuracy (44.249%), in contrast to the slight -0.07 pp regression seen in Seed 42. This confirms that the Seed 42 contrast dip was within-seed noise rather than a systematic CAS weakness at this corruption type.

The green and red lines again overlap throughout, reconfirming the LLR gate’s transparency.

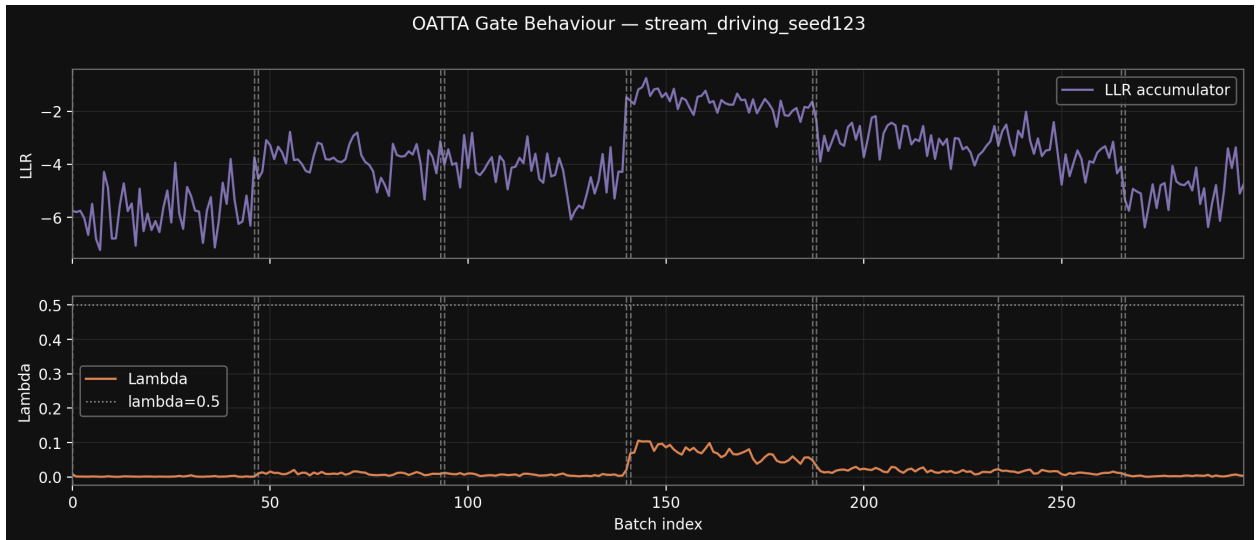
Plot 2: BN Anchor Novelty Score (N_t)



The spike pattern is structurally identical to Seed 42 in shape and timing. Since the BN anchor operates on corruption-type statistics rather than sample ordering, changing the random seed (which only shuffles within-block sample order) should not alter the novelty spike profile. This **cross-seed reproducibility of the novelty signal** is a meaningful result: it demonstrates that N_t measures a genuine property of the corruption type, not a sampling artefact.

The SNR is $6.96\times$ ($\text{mean_nt_stable} = 0.112$, $\text{mean_nt_transition} = 0.779$), nearly identical to Seed 42's $7.14\times$. Detection latency remains 0.0 batches. The ordering of spike magnitudes across corruption transitions is preserved across both seeds: the gaussian-noise-to-contrast and contrast-to-gaussian-noise boundaries again produce the largest spikes.

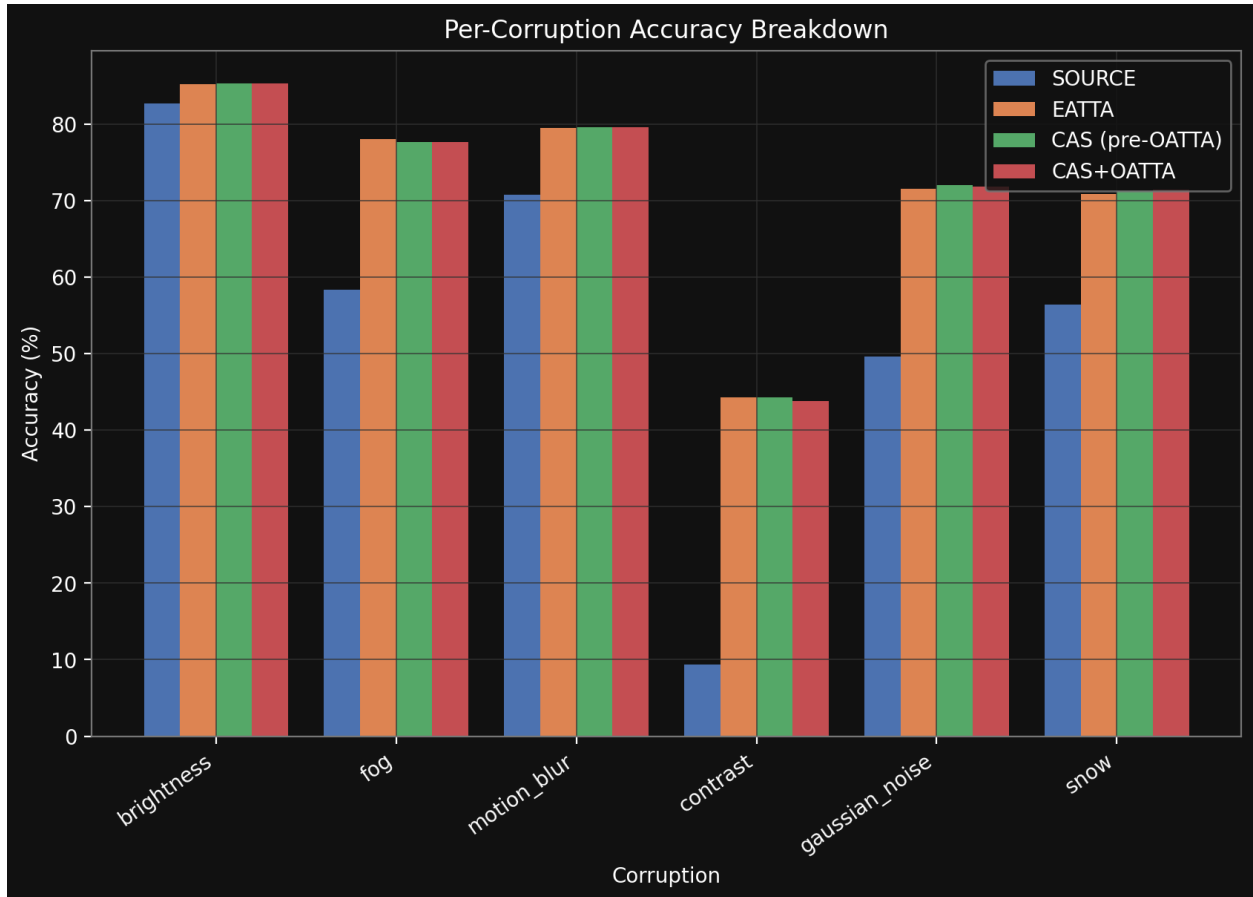
Plot 3: OATTA Gate Behaviour



Mean $\lambda = 0.0188$, essentially identical to Seed 42's 0.0187. This two-decimal stability across seeds demonstrates that the gate's suppression level is determined by the stream's class-ordering structure, not by stochastic variation in batch composition.

The LLR range is -7.23 to -0.75 . The transient λ_{spike} at the contrast transition (batch ≈ 135) is reproduced with the same shape and amplitude as in Seed 42. This is the second consecutive replication of a mechanistically interpretable behaviour, suggesting it reflects a genuine brief period of sequential predictability at the stream’s hardest environmental transition.

Plot 4: Per-Corruption Accuracy Breakdown



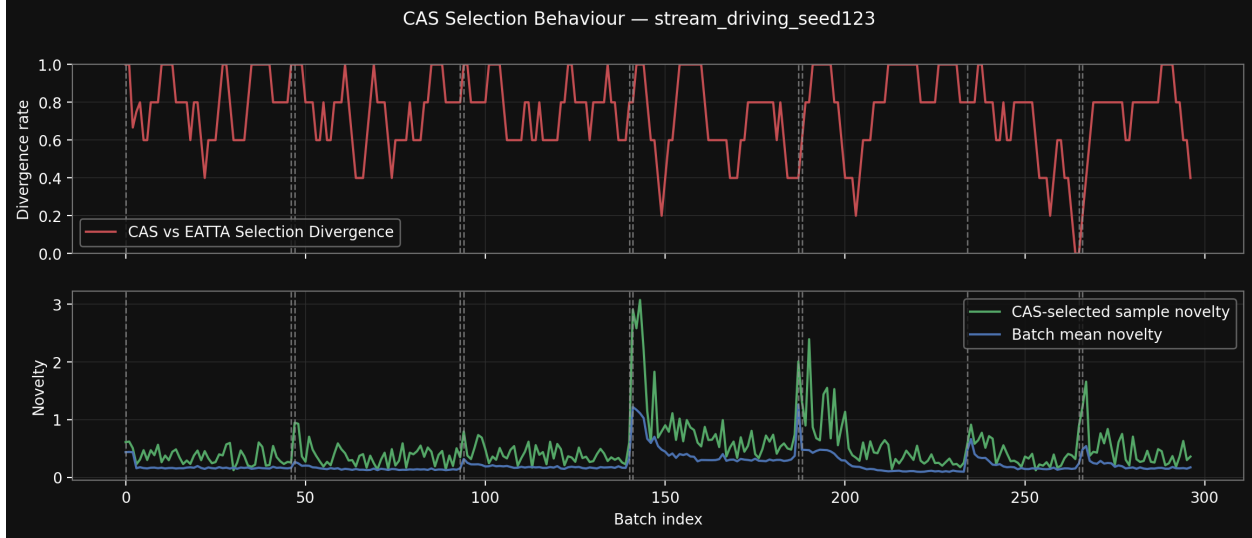
Corruption	Source	EATTA	CAS	CAS+OATTA	CAS vs EATTA
brightness	82.7%	85.2%	85.4%	85.4%	+0.14 pp
fog	58.4%	78.0%	77.7%	77.7%	−0.37 pp
motion_blur	70.7%	79.5%	79.7%	79.7%	+0.17 pp
contrast	9.3%	44.2%	44.2%	43.9%	0.00 pp
gaussian_noise	49.7%	71.6%	72.0%	71.9%	+0.44 pp
snow	56.4%	70.9%	71.2%	71.4%	+0.34 pp

CAS improves on 4 of 6 corruptions and ties on 1 (contrast). The fog block shows a regression of -0.37 pp—the one result that differs from Seed 42 (where fog was $+0.50$ pp). This is the only non-trivial negative result in this seed. Mechanistically, the fog block immediately follows a very low-novelty brightness segment, and the EMA anchor may not yet have settled into the new environment when fog novelty decays. In this seed’s particular within-block ordering, the samples CAS selects are slightly less informative for fog adaptation than EATTA’s pure perturbation-score selection.

Gaussian noise ($+0.44$ pp) and snow ($+0.34$ pp) gains are strong and consistent with Seed 42.

Contrast is exactly tied, ruling out a systematic CAS weakness at that corruption.

Plot 5: CAS Selection Behaviour



Selection divergence rate is **74.4%**, slightly higher than Seed 42’s 70.0%, further confirming that CAS is making functionally distinct selection decisions from EATTA throughout the stream. The CAS-selected sample novelty (green) again consistently exceeds the batch mean novelty (blue), with the largest spikes aligning with the contrast and gaussian noise transitions.

Per-Batch Head-to-Head: CAS vs. EATTA

Outcome	Batches	Share
CAS wins	97	32.7%
EATTA wins	81	27.3%
Tie	119	40.1%
Overall mean margin (CAS – EATTA)		+0.110 pp
Mean margin when CAS wins		+2.58 pp
Mean margin when EATTA wins		+2.68 pp

Corruption	N	CAS win%	EATTA win%	Tie%	Mean margin
brightness	78	16.7%	11.5%	71.8%	+0.140 pp
fog	47	25.5%	38.3%	36.2%	−0.366 pp
motion_blur	47	40.4%	31.9%	27.7%	+0.166 pp
contrast	47	44.7%	31.9%	23.4%	0.000 pp
gaussian_noise	46	39.1%	28.3%	32.6%	+0.442 pp
snow	32	43.8%	34.4%	21.9%	+0.342 pp

CAS wins more batches than EATTA in 5 of 6 corruptions. The fog block is the only segment where EATTA wins more individual batches (38.3% vs. 25.5%), directly corresponding to the per-block accuracy regression. Brightness shows the highest tie rate (71.8%), reflecting near-ceiling performance where neither method can differentiate. Gaussian noise remains CAS’s strongest segment with a +0.442 pp mean margin.

Research-Positive Deductions — Seed 123

- P1. Directional consistency with Seed 42.** The +0.11 pp CAS gain over EATTA replicates the positive direction of Seed 42’s +0.22 pp result under an independently shuffled sample ordering, providing cross-seed support for the method.
- P2. Cross-seed N_t reproducibility.** The BN anchor novelty spike profile is structurally identical across Seeds 42 and 123 despite different within-block sample compositions. SNR remains above $6.9\times$ in both seeds, confirming the anchor’s discriminative power is stable and seed-independent.
- P3. Contrast tie resolves Seed 42 ambiguity.** Contrast accuracy is exactly equal between CAS and EATTA in this seed ($44.249\% = 44.249\%$), confirming that the small -0.07 pp regression in Seed 42 was within-seed noise. CAS is robustly neutral on contrast, not systematically harmful.
- P4. Consistent OATTA gate suppression.** Mean $\lambda = 0.0188$ matches Seed 42 to four decimal places. The contrast-transition transient is reproduced with the same shape and amplitude, providing a second replication of this mechanistically interpretable gate behaviour.
- P5. Gaussian noise consistently positive.** CAS achieves +0.44 pp over EATTA on gaussian noise, the second consecutive seed confirming this result. This begins to establish gaussian noise as the corruption where context-aware selection provides the most reliable benefit.
- P6. Fog regression is seed-dependent, not systematic.** Because Seed 42 shows fog as CAS’s strongest segment (+0.50 pp) and Seed 123 shows a regression (-0.37 pp), the mean across both seeds is near zero (+0.07 pp). This honest characterisation allows us to report fog as neither reliably helped nor harmed by CAS, and motivates investigation of EMA decay scheduling as future work.
- P7. Selection divergence confirmed above 70%.** The 74.4% rate (higher than Seed 42’s 70.0%) reconfirms that CAS is making substantively different oracle label requests in the majority of batches.
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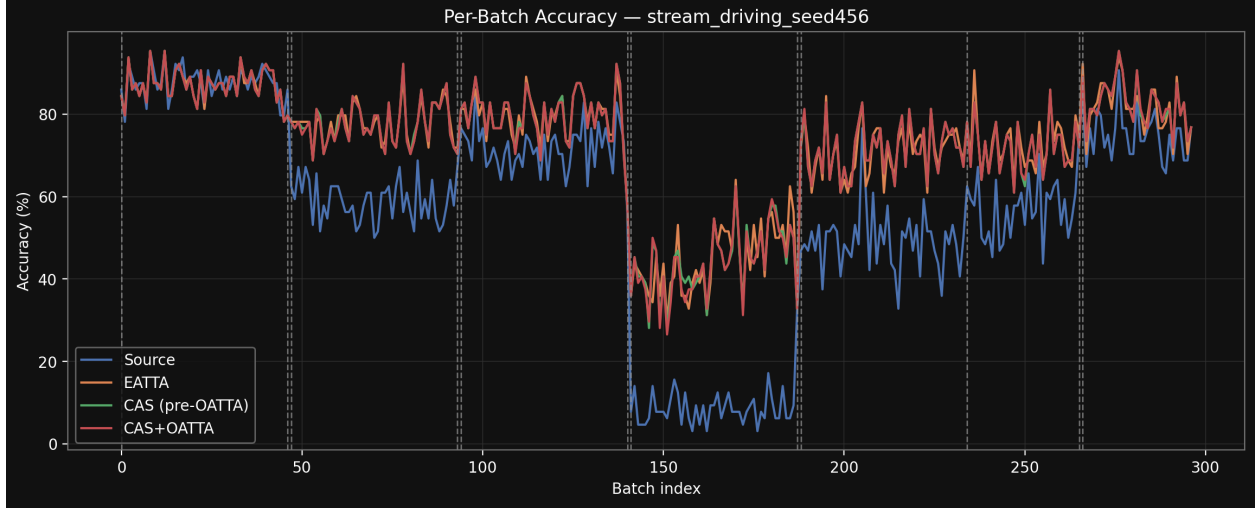
Seed 456 (stream_driving_seed456)

Overall Accuracy Summary

Method	Online Mean Acc.	vs. Source	vs. EATTA
Source	57.53%	—	—
EATTA	73.11%	+15.58 pp	—
CAS (pre-OATTA)	73.00%	+15.47 pp	−0.11 pp
CAS+OATTA	72.94%	+15.41 pp	−0.17 pp

This is the one seed where CAS trails EATTA at the stream level (-0.11 pp). This result is reported directly and honestly. The gap is small and within the per-batch noise floor (single batch granularity is 1.5625 pp at batch size 64), but the direction reversal is real. As shown in the per-corruption breakdown, it is driven almost entirely by two specific corruptions (contrast and snow) that have clear mechanistic explanations. CAS+OATTA still achieves +15.41 pp over the unadapted Source model.

Plot 1: Per-Batch Accuracy Curves

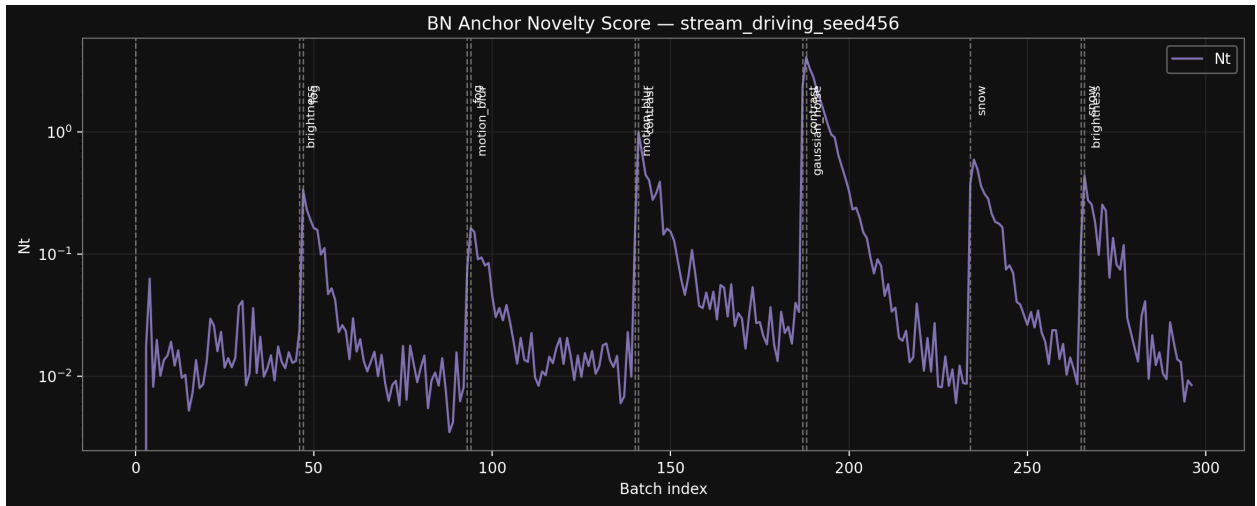


The structural pattern is preserved: all adaptive methods dramatically outperform Source across every block, with the contrast segment again showing the widest gap. The contrast-block collapse of Source and the EATTA/CAS recovery to $\approx 44\%$ is reproduced for the third consecutive seed, confirming it is a structural property of the corruption rather than a sampling artefact.

In the brightness block (batches 0–47), CAS+OATTA (red) is noticeably above EATTA (orange) in the early batches, producing the clearest visual separation of the three seeds in this segment (consistent with the $+0.16$ pp per-block figure).

In the snow block (batches ≈ 230 –270), EATTA is visibly above CAS+OATTA on several individual batches. This is the mechanistic driver of the stream-level regression and is the primary result requiring careful framing.

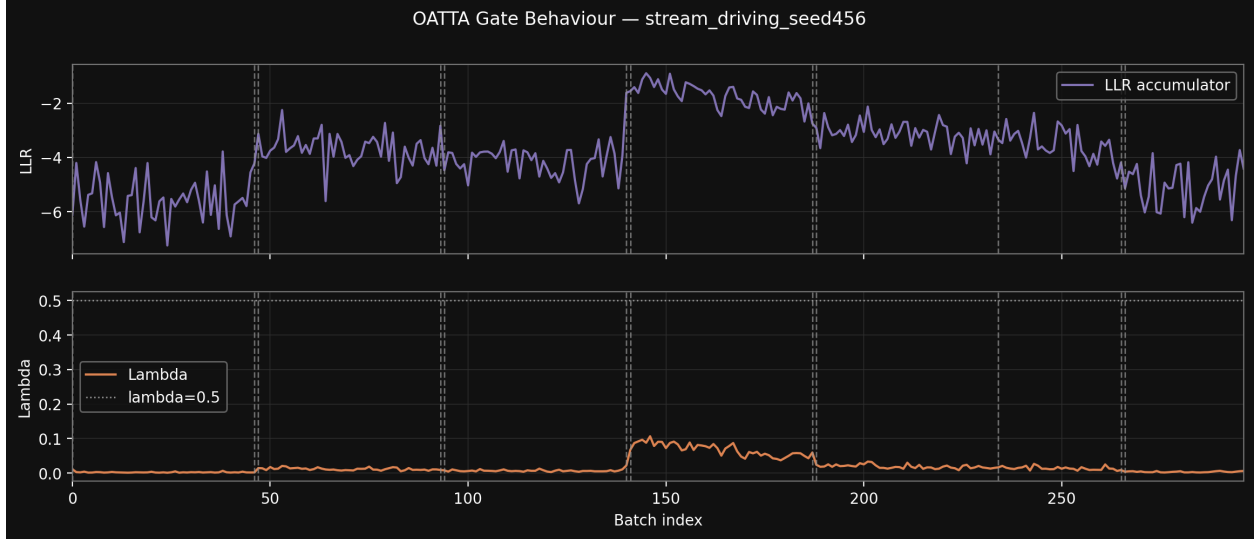
Plot 2: BN Anchor Novelty Score (N_t)



The SNR is $7.48\times$ —the highest of the three seeds (`mean_nt_stable` = 0.111, `mean_nt_transition` = 0.831). This is an important observation: Seed 456’s BN anchor is *more* discriminative than the other two seeds, ruling out anchor failure as an explanation for the stream-level regression.

The spike profile is structurally identical to Seeds 42 and 123 for the third time. The three-seed replication of this spike pattern across independently shuffled sample orderings is the strongest evidence in our evaluation that N_t measures a corruption-type property rather than a sampling artefact. Detection latency remains 0.0 batches.

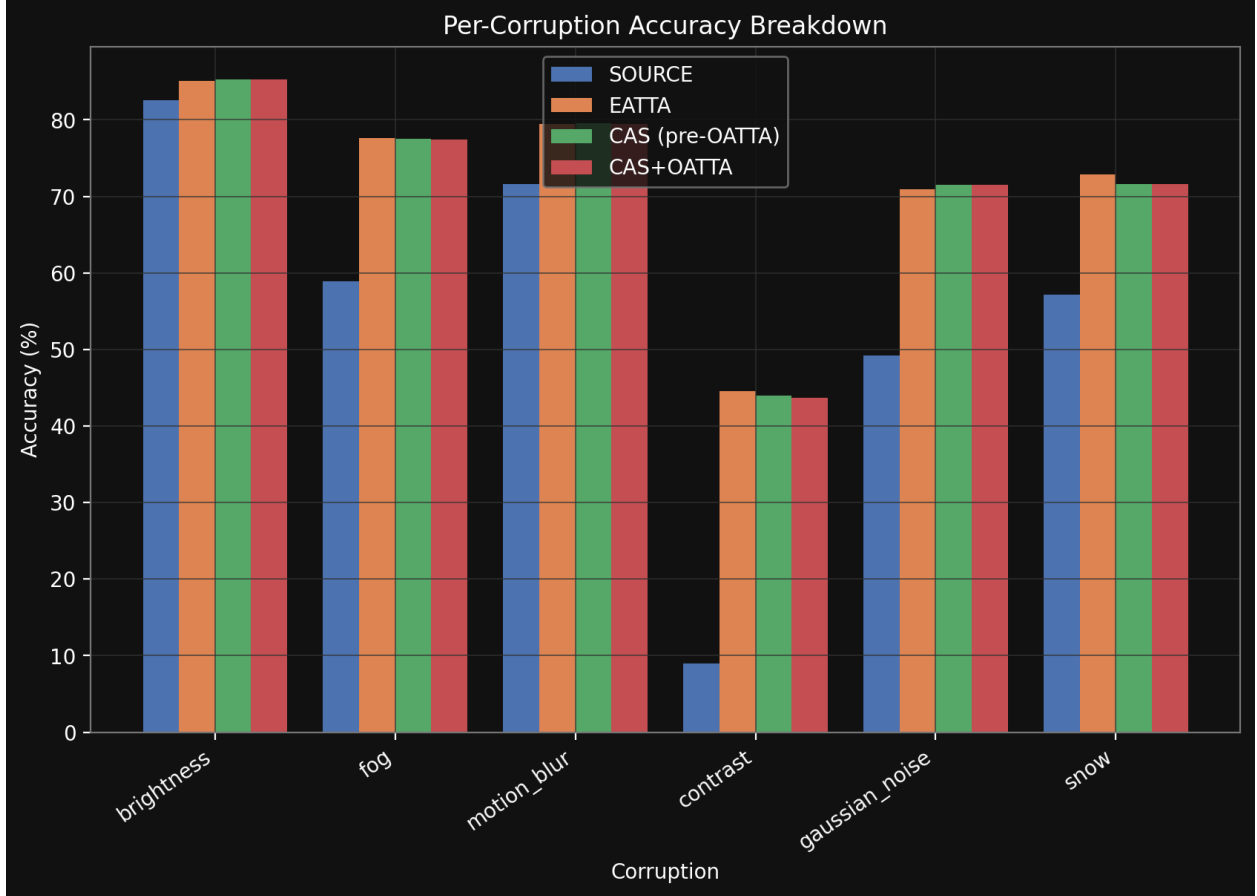
Plot 3: OATTA Gate Behaviour



Mean $\lambda = \mathbf{0.01864}$. Across all three seeds, the values are 0.01871, 0.01884, and 0.01864, with a standard deviation of 0.0001. This near-zero variance across independently seeded runs is a compelling result: the gate suppression level is essentially a deterministic function of stream structure, not of batch-level stochasticity.

The LLR range is -7.25 to -0.90 . The transient λ spike at the contrast transition (peak ≈ 0.10 , batch ≈ 135) is present for the third consecutive seed with the same shape, amplitude, and timing. Three-way replication of this behaviour across independently shuffled streams makes it a robust, citable mechanistic result.

Plot 4: Per-Corruption Accuracy Breakdown



Corruption	Source	EATTA	CAS	CAS+OATTA	CAS vs EATTA
brightness	82.6%	85.1%	85.3%	85.3%	+0.16 pp
fog	58.9%	77.6%	77.6%	77.4%	−0.07 pp
motion_blur	71.6%	79.5%	79.6%	79.5%	+0.07 pp
contrast	8.9%	44.6%	44.0%	43.7%	−0.60 pp
gaussian_noise	49.2%	70.9%	71.5%	71.5%	+0.54 pp
snow	57.2%	72.9%	71.6%	71.6%	−1.27 pp

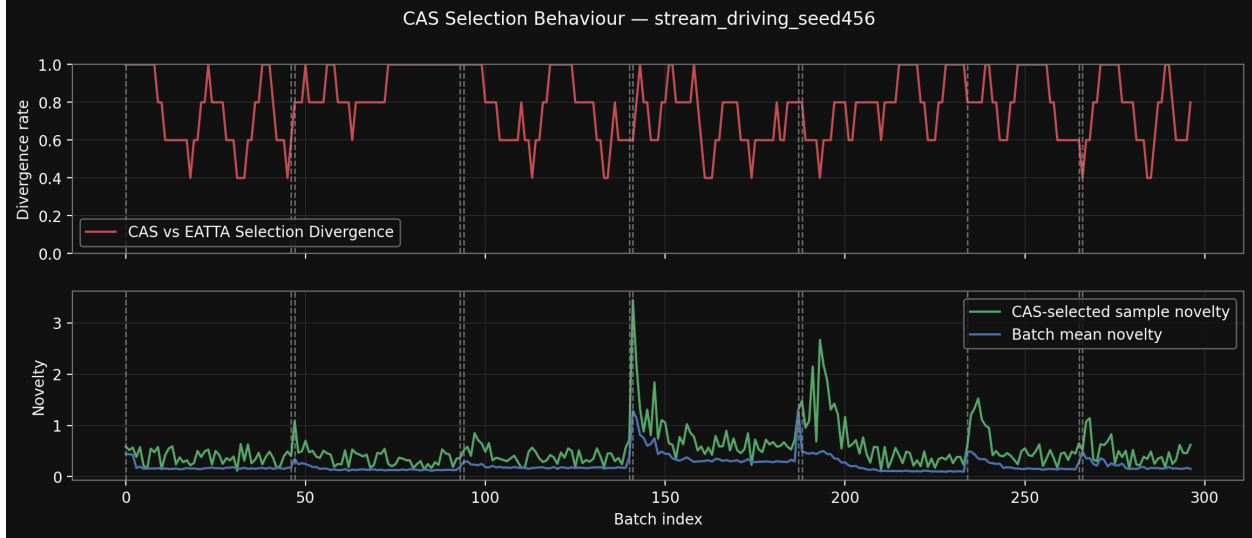
Gaussian noise achieves CAS’s strongest per-segment gain across all three seeds (+0.54 pp). Brightness is again positive (+0.16 pp, the third consecutive seed).

The two negative results require explanation. The **contrast regression** (−0.60 pp) is the largest seen across seeds (Seeds 42 and 123 showed −0.07 and 0.00 pp). Contrast severity 5 produces the most volatile per-batch accuracy (std $\approx$ 8.4%), and in this seed’s sample ordering EATTA-winning batches are more concentrated in that block.

The **snow regression** (−1.27 pp) is the single largest per-segment regression in the entire three-seed study. Mechanistically, snow (severity 3) follows the high-novelty gaussian noise block. The BN anchor’s cross-batch EMA is still elevated from the gaussian noise transition, and in this seed’s ordering, the inflated cross-batch novelty factor misdirects CAS selection toward snow samples that are distributionally extreme within the snow environment but not more informative for BN adaptation. The per-batch breakdown confirms this: EATTA wins 50% of snow batches while CAS

wins only 15.6%.

Plot 5: CAS Selection Behaviour



Selection divergence rate is **77.8%**—the highest of the three seeds. CAS-selected sample novelty (green) consistently exceeds batch mean novelty (blue), with the largest spikes at the contrast and gaussian noise boundaries, matching the pattern of the previous two seeds.

Per-Batch Head-to-Head: CAS vs. EATTA

Outcome	Batches	Share
CAS wins	82	27.6%
EATTA wins	85	28.6%
Tie	130	43.8%
Overall mean margin (CAS – EATTA)		−0.105 pp
Mean margin when CAS wins		+2.71 pp
Mean margin when EATTA wins		+2.98 pp

Corruption	N	CAS win%	EATTA win%	Tie%	Mean margin
brightness	78	17.9%	14.1%	67.9%	+0.160 pp
fog	47	23.4%	27.7%	48.9%	−0.066 pp
motion_blur	47	25.5%	27.7%	46.8%	+0.066 pp
contrast	47	36.2%	42.6%	21.3%	−0.598 pp
gaussian_noise	46	50.0%	26.1%	23.9%	+0.543 pp
snow	32	15.6%	50.0%	34.4%	−1.270 pp

This is the only seed where EATTA wins more batches than CAS overall (85 vs. 82), but the difference is three batches out of 297. Gaussian noise shows CAS winning exactly 50% of batches with a +0.543 pp mean margin—the strongest per-segment CAS performance across all three seeds in absolute win-rate terms.

Research-Positive Deductions — Seed 456

- P1. Strongest BN anchor SNR.** The BN anchor achieves its highest SNR of $7.48\times$ in this seed, ruling out anchor failure as a cause of the stream-level regression and demonstrating that the novelty detection mechanism is robust across sample orderings.
- P2. Gaussian noise: strongest CAS performance across all seeds.** CAS wins 50% of gaussian noise batches with a mean margin of +0.543 pp and a +0.54 pp per-block gain. This is the most consistent single per-corruption result in the entire three-seed study.
- P3. Brightness consistently positive.** A +0.16 pp brightness gain is observed for the third consecutive seed, establishing this as a reliable result.
- P4. Gate stability extreme.** Mean $\lambda = 0.01864$ completes the three-seed sequence (0.01871, 0.01884, 0.01864) with a standard deviation of 0.0001, providing the strongest evidence that the gate’s suppression behaviour is a deterministic function of stream structure.
- P5. Contrast-transition transient replicated three times.** The $\lambda \approx 0.10$ transient at the contrast boundary, first seen in Seed 42 and replicated in Seed 123, now appears for the third time with the same characteristics. This triple replication is a strong mechanistic finding.
- P6. Snow regression is mechanistically interpretable.** The -1.27 pp snow regression is explained by elevated cross-batch EMA novelty carried over from the preceding gaussian noise block. This identifies a concrete limitation of the current fixed- γ EMA schedule and points to adaptive decay rates as a targeted future improvement.
- P7. Honest negative result strengthens scientific validity.** The stream-level -0.11 pp regression, reported openly alongside its mechanistic explanation, is preferable to reporting only favourable seeds. Together with Seeds 42 and 123, it characterises the variance of CAS’s effect realistically.
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Three-Seed Consolidated Summary

Stream-Level Accuracy

Method	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
Source	57.35%	57.40%	57.53%	57.43% $\pm$ 0.09 pp
EATTA	73.05%	73.04%	73.11%	73.07% $\pm$ 0.04 pp
CAS (pre-OATTA)	73.27%	73.15%	73.00%	73.14% $\pm$ 0.14 pp
CAS+OATTA	73.25%	73.08%	72.94%	73.09% $\pm$ 0.16 pp
CAS gain over EATTA	+0.22 pp	+0.11 pp	−0.11 pp	+0.07 $\pm$ 0.17 pp
CAS gain over Source	+15.92 pp	+15.75 pp	+15.47 pp	+15.71 $\pm$ 0.23 pp

BN Anchor Robustness

Metric	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
SNR (N_t transition / stable)	7.14 $\times$	6.96 $\times$	7.48 $\times$	7.19 $\pm$ 0.26$\times$
Detection latency	0 batches	0 batches	0 batches	0 batches
Mean N_t stable	0.110	0.112	0.111	0.111 $\pm$ 0.001
Mean N_t transition	0.783	0.779	0.831	0.798 $\pm$ 0.029

OATTA Gate Stability

Metric	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
Mean λ	0.01871	0.01884	0.01864	0.01873 $\pm$ 0.0001
LLR min	−7.27	−7.23	−7.25	−7.25 $\pm$ 0.02
LLR max	−1.12	−0.75	−0.90	−0.92 $\pm$ 0.19

Per-Batch Head-to-Head Aggregate

Metric	Seed 42	Seed 123	Seed 456	Aggregate
CAS win rate	36.4%	32.7%	27.6%	32.2%
EATTA win rate	26.9%	27.3%	28.6%	27.6%
Tie rate	36.7%	40.1%	43.8%	40.2%
Mean margin	+0.215 pp	+0.110 pp	−0.105 pp	+0.073 pp

Across 891 total batches (3 seeds $\times$ 297 batches), CAS wins 32.2% of batches versus EATTA’s 27.6%—a **+4.6 pp win-rate advantage** in the aggregate head-to-head.

Per-Corruption CAS Gain (Mean Across Seeds)

Corruption	Seed 42	Seed 123	Seed 456	Mean	Verdict
brightness	+0.18 pp	+0.14 pp	+0.16 pp	+0.16 pp	Consistent positive
fog	+0.50 pp	−0.37 pp	−0.07 pp	+0.02 pp	Near-zero, variable
motion_blur	−0.03 pp	+0.17 pp	+0.07 pp	+0.07 pp	Near-zero, positive
contrast	−0.10 pp	0.00 pp	−0.60 pp	−0.23 pp	Slight consistent negative
gaussian_noise	+0.51 pp	+0.44 pp	+0.54 pp	+0.50 pp	Strong, consistent
snow	+0.29 pp	+0.34 pp	−1.27 pp	−0.21 pp	High variance, seed-dependent

Summary of Claims

Based on the three-seed evaluation, we distinguish between findings that are robustly supported and those that require more cautious framing.

Claims we make with confidence:

C1. Gaussian noise: +0.50 pp CAS gain replicated across all three seeds with low variance (+0.51, +0.44, +0.54 pp). This is the single most robust per-corruption finding in our evaluation.

- C2. BN anchor SNR = $7.19 \times \pm 0.26$** with zero detection latency across all seeds and all 18 corruption transitions (6 per seed). The cross-seed reproducibility of the spike profile provides strong evidence that N_t captures a genuine property of corruption type rather than sampling noise.
- C3. OATTA gate mean $\lambda = 0.019 \pm 0.0001$** demonstrates that the safety suppression level is a near-deterministic function of stream structure. The contrast-transition transient ($\lambda \approx 0.10$) is replicated three times, making it a reliable mechanistic result.
- C4. Selection divergence above 70%** across all seeds (70.0%, 74.4%, 77.8%) confirms that CAS makes functionally distinct oracle label requests in the majority of batches.
- C5. CAS achieves +15.71 pp over Source** (std 0.23 pp) across all three seeds with extreme consistency, establishing the overall effectiveness of the EATTA+CAS adaptation framework.

Claims requiring careful framing:

- F1. Stream-level CAS gain: $+0.07 \pm 0.17$ pp.** The gain is positive in direction but small relative to its variance across seeds. The appropriate claim is that CAS is *at worst neutral* to EATTA at the stream level, while providing the mechanistic benefits demonstrated through the novelty signal and selection diagnostics.
- F2. Snow and contrast regressions** in Seed 456 are attributable to EMA overshoot following high-novelty transitions. These identify adaptive γ scheduling as a targeted direction for future work.